

Vijayaragavan S

FULL STACK DEVELOPER

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Computer Science undergraduate graduating in 2027, building applications with React, TypeScript, Node.js and PostgreSQL. Experience spans authentication, real-time messaging, data dashboards and AWS deployment, alongside an offline Android research app and an IEEE publication.

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript, Vite, HTML, CSS, Tailwind CSS

Backend and data: Node.js, Express.js, Flask, REST APIs, Socket.IO, PostgreSQL, Prisma, SQL

Deployment: AWS (EC2, S3, CloudFront, SES), Docker, Docker Compose, Nginx, PM2

Additional tools: Python, Java, Git, Postman, pytest, Streamlit, Pandas, Plotly

PROJECTS

CampusRent | Campus Rental Marketplace

2025-Present | React, TypeScript, Node.js, PostgreSQL, AWS | Live application

- Built a rental marketplace with listings, booking workflows, reviews and separate student, landlord and admin access; added real-time chat with Socket.IO.
- Designed the Prisma/PostgreSQL data model and Express APIs; implemented JWT authentication, request sanitization and configurable API rate limiting.
- Added image uploads to S3 with a 5 MB file limit and SES email verification; deployed the frontend on S3/CloudFront and the backend on EC2 with Nginx and PM2.
- Configured Docker Compose for the database and application services, with a PostgreSQL health check and Prisma migrations at backend startup.

AnaemiaScreen | Research App and Data Collection

2026 | Java, Python, Streamlit, PyTorch, LiteRT

- Built a Streamlit screening demo and an offline Android app with camera/gallery input, automated conjunctiva cropping and calibrated model outputs.
- Added participant records, image capture, CSV tables and ZIP export; kept model outputs hidden during data collection to avoid influencing the collector.
- Validated the Java/Python processing pipeline with zero decision differences across **634 scored** cases; added resolution checks and a low-resolution model route.

Carbon Emissions Intelligence Platform

2025-Present | Python, Flask, PostgreSQL, Pandas, Plotly, Docker | Source code

- Built a Flask API and PostgreSQL data layer for carbon accounting; added ingestion, validation and transformation workflows for emissions records.
- Created interactive dashboards for emissions trends and forecasts, and containerized services for repeatable local setup and deployment.

PUBLICATION

Vijayaragavan Sivasamy et al. (first author). Real-Time Pothole Detection Using Dual Accelerometer Sensors with Temporal Convolutional Networks and Attention Mechanisms. IEEE conference publication, 2026.

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EDUCATION

B.Tech, Computer Science and Engineering | 2023-2027 (Expected)

SRM Institute of Science and Technology, Chennai | **CGPA: 9.46/10**

Relevant coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

Certifications: Java - NPTEL | SAP Backend ABAP Cloud Certificate